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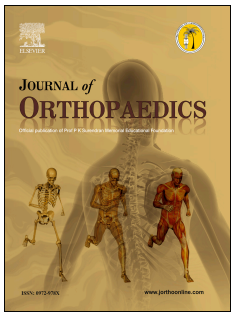

Fantasy football points capture performance declines in National Football League offensive skill players following an ankle injury

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Fantasy Football Points Capture Performance Declines in National Football League Offensive Skill Players Following an Ankle Injury

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Running Title: Performance Declines in NFL Offensive Players After Ankle Injury

The authors declare no conflicts of interest. Ethical approval was not sought for the present study because all data was obtained from a publicly available database.

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ABSTRACT

Background: The ankle is one of the anatomic sites most frequently injured in National Football League (NFL) players. Ankle injuries have previously been shown to have long-lasting negative impacts, and have been associated with impaired athletic performance. The aim of this study was to use fantasy football points as a metric to evaluate the impact of ankle injuries on NFL offensive skill player performance.

Methods: An open-access online database was used to identify NFL players who sustained ankle injuries from 2009-2020. Another public online database was used to determine fantasy points and other performance metrics for injured offensive skill players in the seasons before and after their ankle injury. Injured players were matched to a healthy control by position, age, and BMI. Paired T-tests were performed to evaluate performance metrics before and after the ankle injury. An ANCOVA was performed to assess the effect of return to play (RTP) time and injury type on fantasy performance.

Results: 303 players with ankle injuries were included. Fantasy output, including average points per game (PPG) and total fantasy points accrued in one season, significantly decreased in the season following a player's ankle injury ($p<0.0001$). In running backs, tight ends, and wide receivers, performance significantly decreased in every metric evaluated ($p<0.0001$). In quarterbacks, there was no significant change in performance, except for a decrease in the number of games played ($p=0.0033$) and in the number of interceptions thrown ($p=0.029$).

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Conclusion: Assessing fantasy football output revealed a decrease in player performance in the season following an ankle injury, especially in route-running players. These results can be used to inform injury prevention and rehabilitation practices in the NFL.

KEYWORDS: Sports, American Football, Ankle Sprain, Performance Outcomes

INTRODUCTION

Ankle injuries represent 10-28% of all sports-related traumas, and can cause lingering problems for athletes.[1,2] Pain and instability are the most prevalent concerns, followed by crepitus, weakness, stiffness, and swelling.[2] The nature of the residual problems can also depend on injury frequency. For ankles sprained under and over 5, the primary concerns are pain and instability, respectively.[2] Patients who sustain an inversion ankle injury during sports experienced persistent symptoms for at least two years after the initial injury.[3] Additionally, ankle injuries have been associated with altered proprioception, further affecting athlete performance and recovery.[4]

American football is one of the most high-risk sports in the United States.[5] Players in offensive positions and “skill positions” are especially vulnerable to foot and ankle injuries due to the intense force and torque exerted on the distal extremities during actions like running, cutting, and tackling.[6–8] Offensive linemen, running backs, and tight ends exhibit higher overall rates of ankle injuries, while special teams players demonstrate the lowest incidence.[6,7]

Despite the significant impact of ankle injuries in the NFL, no studies have evaluated the effects of these injuries on football performance in terms of fantasy football output.[9] Fantasy football points are an underutilized resource for this purpose, as they combine many aspects of a

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player's performance into a single representative value that facilitates comparison. Our study utilizes fantasy football points as a proxy measure for offensive skill player performance after ankle injury.[10] We hypothesize that fantasy points will reflect decreases in player performance.

METHODS

Data Source

This study was a retrospective review of an open-access, online database called Pro Football Reference, which utilizes the official statistics partner of the NFL, Sportradar.[11] Because the database is publicly available, Institutional Review Board approval was not necessary. Pro Football Reference keeps a database of all player injuries published on the NFL official injury report since 2009. Many peer-reviewed studies have utilized this injury database.[9,10,12–14] We recorded each ankle injury in the database, as well as how many respective games were missed. Using reputable sports news websites as confirmation, ankle injuries were classified as an ankle sprain, high ankle sprain, bone bruise, ligament tear, Achilles tendon rupture, ankle fracture, or ankle dislocation. To avoid a confounding effect from the varying injury severity, we only included ankle sprains, high ankle sprains, and bone bruises.

Players were excluded if they experienced more than one injury in a given season, and players with multiple ankle injuries were only counted for their first ankle injury. Players were also excluded if they played less than three games or accrued exactly zero fantasy points in the season before their injury, and if they did not return to play following their injury. Other information collected from Pro Football Reference included date of injury, return to play (RTP) time, position, age, and body mass index (BMI).

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Fantasy football points are used in this study as a proxy for measuring player performance. Using fantasy points is a valuable way to assess offensive skill player performance because it incorporates many aspects of a player's on-field statistics into a single numerical score that is easily comparable across players of the same position. For example, for wide receivers, rather than only using receiving yards as a measure of performance, using their fantasy point total also allows us to assess rushing yards, number of catches, number of touchdowns, and more categories that not illustrate the player's skill, but also to the player's importance and utilization in the offensive scheme. Player statistics for each individual performance measure factored into the fantasy point scores are also separately detailed, to show the differences in player performance in more concrete, validated terms (**Tables 3 and 4**).

In the standard scoring system for fantasy football, certain point values are ascribed to different performance metrics, outlined in **Supplemental Table 1**. Individual players can accrue fantasy points only if they are an offensive skill player – namely a quarterback, running back, tight end, or wide receiver – or if they are a kicker. All other players were excluded from our analysis. Kickers were also excluded from our analysis because only four kickers reported ankle injuries during the study period.

Fantasy points and their corresponding performance metrics were collected from the open-access online database FantasyData, which provides sports research tools and fantasy football data.[15] FantasyData has been previously been cited as a reliable source with peer-reviewed approval.[10]

Outcome Measures

Performance metrics collected from quarterbacks for each season included number of games played, passing yards, passing touchdowns, interceptions, rush yards, rush touchdowns,

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average fantasy points scored per game, and total fantasy points scored. Performance metrics collected from route runners (including running backs, tight ends, and wide receivers) for each season included number of games played, rush yards, number of receptions, receiving yards, total touchdowns, average fantasy points scored per game, and total fantasy points scored.

Statistical Analysis

Statistical analysis was conducted using Microsoft Excel (Microsoft corporation, Richmond, WA) and RStudio (RStudio: Integrated Development for R. RStudio, PBC, Boston, MA). Performance metrics were collected for the seasons immediately before and after the one in which a given player experienced an ankle injury. Paired T-tests were performed to evaluate the difference in performance before and after the year of the ankle injury. An analysis of covariance (ANCOVA) was performed to determine the effect of RTP time and injury type on performance in the season following a player's ankle injury. A post-hoc Bonferroni analysis was conducted to evaluate if specific injury types had varying impacts on performance in the season following the injury. A P value of 0.05 or less was used to determine statistical significance.

A matched cohort analysis was performed. Injured players were matched to healthy controls 1:1 by position, age, and BMI, in order of priority. Paired T-tests were performed to evaluate differences in performance in the seasons before and after the year of the matched player's ankle injury. Independent, two-tailed T-tests were performed to evaluate differences in demographics, including age and BMI.

RESULTS

Player Demographics

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Over the course of the 2009-2010 to 2019-2020 seasons, 941 players of 5,536 total sustained an ankle injury. Of the 303 offensive skill players who sustained an ankle sprain, a high ankle sprain, or an ankle bone bruise during the study period, 17 were quarterbacks, 106 were running backs, 55 were tight ends, and 125 were wide receivers. Of the ankle injuries included in the study, 214 were classified as sprains, 87 were classified as high ankle sprains, and 2 were classified as bone bruises. Each injured player was matched to a healthy control by their position, age, and BMI. The mean ages of the injured cohort ("cases") and the uninjured cohort ("controls") were 26.59 (SD 3.02) and 28.77 (SD 2.33), respectively. The mean BMIs of the case and control cohorts were 26.76 (SD 2.23) and 28.72 (SD 2.25), respectively. These demographic differences between the two cohorts were not statistically significant, with P values of 0.089 and 0.93 for age and BMI, respectively.

Overall Performance Metrics

Offensive skill players who sustained ankle injuries during the study period showed statistically significant decreases in number of games played, average fantasy points scored per game (PPG), and total fantasy points in the season after their injury. The matched controls did not show statistically significant decreases in these measures. These results are detailed in **Table 1**. Factoring in RTP and injury type as covariates did not significantly impact average fantasy PPG and total fantasy points per season. However, including RTP as a covariate when analyzing number of games played in the seasons before and after the ankle injury increased the P value from <0.0001 to 0.0127. Similarly, including injury type as a covariate increased the P value from <0.0001 to 0.132. This indicates that RTP time and injury type could have confounding influences on the number of games played in the season following an ankle injury. These results

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are detailed in **Table 2**. Post-hoc Bonferroni adjustments showed no significant differences between the different injury types and their respective impacts on number of games played, PPG, or total points after the player's ankle injury. The changes in fantasy points before and after a player's ankle injury are summarized in **Figure 1**.

Route Runner Performance Metrics

Route runners who sustained an ankle injury experienced significantly decreased performance in every metric measured. In comparison, the matched controls did not perform significantly differently after the index year, with the exception of a decrease in receiving yards (p=0.049). The differences in performance metrics are detailed in **Table 3**.

Quarterback Performance Metrics

Quarterbacks who sustained ankle injuries did not exhibit many major differences in performance in the season following said injury. The lone exceptions include decreases in the number of games played and in interceptions thrown in the year following the player's ankle injury. These results are detailed in **Table 4**.

DISCUSSION

In this study of 303 ankle injuries that occurred over the course of 11 NFL seasons, we found that injured NFL players exhibit significant detriments in fantasy football production in the year following their ankle injury. When stratifying by player position, we found that route runners show significantly impaired fantasy football production across all metrics studied, compared to quarterbacks, whose performance is generally unaffected. These results support

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those of previous analyses of fantasy output following lower extremity injury.[16] Our findings not only display an interesting injury pattern, but are also clinically relevant. Athletes, coaches, team medical staff, and athletic trainers may use these results to guide clinical decision-making with regard to return-to-play timing, rehabilitation protocols, and enhanced prevention strategies for athletes with ankle injuries, with the aim of mitigating the decrease in post-injury performance noted in our study.

Previous studies have identified ankle injuries as one of, if not the most common musculoskeletal injury sustained by football players.[8,14,17] In fact, a study of players attending the NFL combine showed that 53.2% of players had a previous ankle injury, even before the player enters the NFL.[6] The prevalence of ankle injuries in NFL players poses a particular challenge in light of evidence that these injuries have consistently been shown to impact players long-term.[14] After ankle injuries, football players are likely to experience persistent symptoms, especially stiffness, and can develop chronic ankle instability (CAI).[18] CAI is characterized by laxity and mechanical instability, leading to insufficiency of the lateral ankle ligament complex that interferes with activity.[19] Onset of CAI can occur after just one ankle sprain and worsens with repetitive injuries, which is concerning when considering that 43% of acute ankle sprains reported in football players are recurrent injuries.[20] An explanation for this trend is that ankle sprains are most often reported in sports characterized by motions such as running, cutting, and jumping: motions that are integral to offensive football players moving the ball downfield.[20] Route runners run with the ball, cut to evade defensive opponents, and jump to catch thrown balls. Considering the integral biomechanical role of the ankle in running football routes, our findings that route runners exhibit decreased performance in every metric during the season following an ankle injury are well-supported.

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We found that quarterbacks did not exhibit the same decrease in performance in the season following an ankle injury that was observed in route runners. These results support previous literature that have also concluded not only that ankle injuries are less frequent in quarterbacks, but also that they do not significantly impact quarterback performance.[14,21] Ankle injuries have the potential to limit quarterbacks' pocket mobility and rushing efficacy, however, these skills are not critical for quarterback performance.[14]

Surprisingly, adjusting for RTP time and injury type did not affect the decreases observed in PPG and total fantasy points. However, they did both exert a confounding effect on the number of games in which the athlete played in the season following their ankle injury, given that increased RTP time and more severe injuries could sideline players continuing into the following season. The fact that players still exhibit significant decreases in fantasy points regardless of ankle injury severity may point to the effectiveness of rehabilitation programs within the NFL. Conversely, because players with longer RTP time and more severe injuries may be sidelined for a period spanning more than one season, it is logical that these factors would exert a confounding effect on the number of games in which that player participates during the season following their ankle injury.

This study should be interpreted in the context of its limitations. Firstly, as this a retrospective study, we can only report associations, not causation. We lacked access to the NFL electronic medical record. Fantasy football points are not a previously validated performance measure. While the on-field components that comprise fantasy football points are direct performance statistics, they are differentially weighted within the fantasy score, therefore some extrapolation is pertinent while interpreting the results. There are inherent limitations that come with using a public injury database, including under- or over-reporting of injuries based solely on

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media reports. Some injuries in the Pro Football Reference database were classified as "other", which could contribute to an undercounting of ankle injuries. This study did not account for ankle injuries sustained prior to the inception of the study period. The Pro Football Reference database does not include information about mechanism of injury, such as contact or non-contact, nor information about playing conditions that could influence injury risk, so those factors were not included in our analysis.

CONCLUSION

Over the course of eleven NFL seasons, there were 332 ankle injuries sustained by offensive skill players. Fantasy output, measured in terms of both average points scored per game and total season points, decreased significantly during the season following an ankle injury. This finding was not influenced by RTP, type of ankle injury, BMI, or player age. Route runner positions experienced significant decreases across all performance metrics evaluated, while quarterbacks did not show notable decreases in specific performance metrics. These results confirm prior studies showing the harmful impact of ankle injuries in football players. Further research should examine NFL rehabilitation protocols, and how they can be optimized to improve player performance upon their return from an ankle injury.

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Tables

Table 1. Fantasy Football Performance of All Offensive Skill Players One Season Before and After an Ankle Injury

	Before Ankle Injury Mean (SD)	After Ankle Injury Mean (SD)	P Value
Cases			
Number of Games Played	13.25 (3.56)	9.89 (6.27)	**<0.0001**
Average Fantasy Points Scored Per Game	6.20 (5.01)	4.77 (4.91)	**<0.0001**
Total Fantasy Points Scored in One Season	85.15 (75.42)	62.97 (73.71)	**<0.0001**
Controls			
Number of Games Played	12.62 (4.20)	12.16 (4.66)	0.12
Average Fantasy Points Scored Per Game	4.63 (4.47)	4.34 (4.20)	0.13
Total Fantasy Points Scored in One Season	63.32 (68.04)	57.52 (62.78)	0.057

Table 2. ANCOVA Adjustment for the Effects of Return to Play Time and Injury Type on Performance

	Unadjusted P Value	P Value Adjusted for RTP	P Value Adjusted for Injury Type
Number of Games Played	**<0.0001**	**0.0127**	0.132
Average Fantasy Points Scored Per Game	**<0.0001**	**<0.0001**	**<0.0001**
Total Fantasy Points Scored in One Season	**<0.0001**	**<0.0001**	**<0.0001**

Table 3. Fantasy Football Performance and Performance Metrics of Route Runners One Season Before and After an Ankle Injury

	Before Ankle Injury Mean (SD)	After Ankle Injury Mean (SD)	P Value
Cases			
Number of Games Played	13.07 (3.91)	10.07 (6.21)	**<0.0001**
Total Season Rush Yards	199.36 (379.41)	147.62 (334.20)	**<0.0001**
Total Season Number of Receptions	31.88 (26.47)	25.83 (27.92)	**<0.0001**

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Total Season Receiving Yards	375.63 (363.46)	288.83 (357.24)	**<0.0001**
Total Season Touchdowns	3.72 (4.16)	2.80 (3.57)	**<0.0001**
Average Fantasy Points Scored Per Game	5.85 (4.77)	4.53 (4.62)	**<0.0001**
Total Fantasy Points Scored in One Season	79.11 (71.60)	59.65 (67.38)	**<0.0001**
Controls			
Number of Games Played	12.73 (4.16)	12.30 (4.57)	0.17
Total Season Rush Yards	113.18 (279.71)	94.12 (239.30)	0.12
Total Season Number of Receptions	26.51 (26.72)	24.66 (25.99)	0.13
Total Season Receiving Yards	311.92 (348.07)	282.11 (326.34)	**0.049**
Total Season Touchdowns	2.80 (3.40)	3.17 (11.80)	0.58
Average Fantasy Points Scored Per Game	4.25 (3.95)	3.95 (3.66)	0.12
Total Fantasy Points Scored in One Season	58.63 (60.11)	52.71 (54.94)	0.054

Table 4. Fantasy Football Performance and Performance Metrics of Quarterbacks One Season Before and After an Ankle Injury

	Before Ankle Injury Mean (SD)	**After Ankle Injury Mean (SD)**	**P Value**
Cases			
Number of Games Played	12.50 (3.99)	9.00 (6.60)	**0.0033**
Total Season Passing Yards	2546.00 (1674.74)	1894.80 (2016.48)	0.053
Total Season Passing Touchdowns	15.00 (13.38)	13.20 (14.94)	0.33
Total Season Interceptions	9.29 (5.61)	5.61 (5.72)	**0.029**
Total Season Rush Yards	96.15 (93.19)	82.67 (89.86)	0.14
Total Season Rush Touchdowns	0.93 (1.00)	0.53 (1.02)	0.14
Average Fantasy Points Scored Per Game	12.08 (6.21)	9.93 (8.62)	0.16

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Total Fantasy Points Scored in One Season	164.67 (111.76)	129.45 (140.96)	0.10
Controls			
Number of Games Played	10.74 (4.51)	9.79 (5.53)	0.19
Total Season Passing Yards	2123.05 (1678.48)	2077.32 (1602.32)	0.86
Total Season Passing Touchdowns	13.53 (12.23)	12.53 (10.50)	0.53
Total Season Interceptions	8.84 (5.70)	6.58 (5.83)	4
Total Season Rush Yards	135.79 (175.03)	149.63 (189.40)	0.67
Total Season Rush Touchdowns	1.68 (3.25)	1.21 (1.78)	0.42
Average Fantasy Points Scored Per Game	11.15 (7.17)	10.94 (6.77)	0.88
Total Fantasy Points Scored in One Season	142.76 (126.31)	139.03 (114.81)	0.82

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Figures

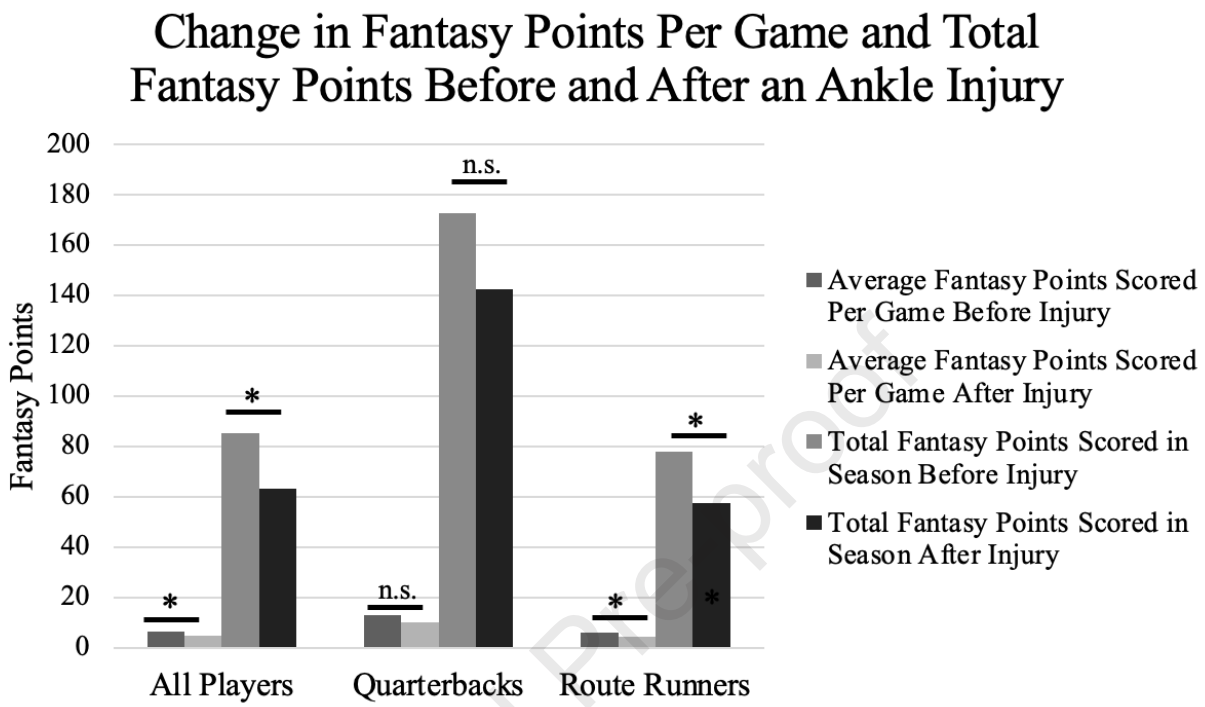

Figure 1. Change in Fantasy Points Per Game and Total Fantasy Points Before and After an Ankle Injury. * indicates $p \leq 0.05$, n.s. indicates a lack of statistical significance.

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This study was a retrospective review of an open-access, online database. Because the database is compiled via publicly available media reports and does not contain specific medical information, Institutional Review Board approval was not necessary.

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